



AGNIESZKA SURTEL

SOFTWARE DEVELOPER



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[GitHub](#)

TECHNICAL SKILLS & TOOLS

Programming Languages:

Python, C++, PHP 8.x, JavaScript

Backend: Node.js, REST APIs, MVC architecture, SQL (PDO, CRUD, Security/SQL Injection prevention), SQL Databases, CRUD Operations, XAMPP / Apache

Frontend: HTML5, CSS3, Chart.js, DOM Manipulation, jQuery

Learning / Expanding to: **C#** (strong interest & readiness to work across languages)

Architecture & Best Practices:

SOLID, Clean Code, TDD, DRY, KISS, OOP

Tools & Environment:

Git, GitHub, VS Code, Visual Studio, XAMPP, Code::Blocks

EDUCATION

Master of Science Eng.

University of Life Sciences
Poland, Lublin, Biotechnology
2016 – 2021

ABOUT ME

As a **M.Sc. in Engineering**, I combine technical knowledge with a holistic approach to software development. I write **clean, testable** code driven by **SOLID** principles, with performance and scalability in mind. Proficient in Python and C++, with a strong readiness to expand my skills into **C#** and desktop architectures to bridge backend logic with modern interfaces.

Stack: **Python, C++, PHP 8.x, JavaScript & more**

INDEPENDENT SOFTWARE DEVELOPMENT & OPEN SOURCE PROJECTS

BESS-control-toolkit – Battery Energy Storage System Control Core

- **Role:** Independent C++ | Python Developer
- **Tech:** C++, Python, State Machine Architecture, Systems Programming
- **Key Achievements:**
 - Developed a robust control core for BESS, ensuring safe transition between operating states (IDLE/CHARGING/SHUTDOWN).
 - Implemented decoupled safety-monitoring logic to prioritize system integrity.
 - Engineered a cross-tool data pipeline (C++ to Python) for automated operational logging and system diagnostics.

BudgetApp – Personal Financial Management Web App

- **Role:** Independent Full-Stack Developer
- **Stack:** PHP 8.x, MySQL, JavaScript (ES6+), MVC, Bootstrap
- **Key Responsibilities & Achievements:**

PROGRAMMING SCHOOL

An intensive, multi-stage software engineering program focused on computer science fundamentals, algorithm design, and professional application
<https://przyszlyprogramista.pl/>

A2B Hobro

Danish language school

2025 – 2026

SOFT SKILLS

- Analytical Thinking & Problem-Solving
- Self-starter mindset
- Collaboration & Teamwork
- Quality-Oriented & Detail-Focused
- Adaptability & Fast Learning

LANGUAGES

- Polish
- English
- Danish

INTERESTS

- AI Tools: Exploring AI-driven development and workflow automation
- FinTech & Personal Finance: Passionate about building tools for financial management.
- Creative Arts: Painting with watercolors and crafting handmade art
- Physical Activity: Regular sports training to maintain focus and discipline.

- Built dynamic **CRUD transaction** management with secure PDO implementation.
- Integrated **MySQL database** and visualized financial data using Chart.js.
- Applied MVC architecture for clean, scalable, and secure codebase.

[View live Budget App](#)

Contact Management Console App – C++ App

- **Role:** Independent C++ Developer
- **Stack:** ++, OOP, File I/O, Git
- **Key Responsibilities & Achievements:**
 - Designed modular **OOP architecture** with clear separation of concerns
 - Integrated persistent storage **using File I/O to save, read, and update** contact credentials.
 - **Structured** the **codebase** into separate header (.h) and source (.cpp) files.

[View demo Contact App](#)

Personal Portfolio Website

- **Role:** Frontend Developer
- **Stack:** HTML5, CSS3, Bootstrap, JavaScript (ES6+), GitHub Pages
- **Key Responsibilities & Achievements:**
 - Developed fully responsive (**RWD**) interface for project showcase
 - Enhanced **user experience** with interactive UI components and CSS3 transitions
 - Deployed and hosted production-ready site via GitHub Pages

[View Portfolio Webside](#)

Interactive Simon Game Web App

- **Role:** Frontend Developer
- **Stack:** JavaScript (ES6+), HTML5, CSS3, Git
- **Key Responsibilities & Achievements:**
 - Developed dynamic game logic using pure vanilla JavaScript.
 - Implemented **DOM manipulation** for real-time user interaction and sequence validation..
 - Engineered algorithmic patterns for randomized game states and **level progression**
 - Integrated synchronized audio-visual effects for an engaging user experience.

[View Simon Game](#)

ADDITIONAL PROFESSIONAL EXPERIENCE

Production Assistant & Vendor Relations | Vebbestrup Flødeis
– Sdr Truevej 1, 9500 Hobro | September 2024 – Present